

Questions with Answer Keys

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Q1 - Single Correct

If for an AP, $a_1, a_2, a_3, \dots, a_n, \dots$ $a_1 + a_3 + a_5 = -12$ and $a_1 a_2 a_3 = 8$, then the value of $a_2 + a_4 + a_6$ equals

- (1) -12
- (2) -16
- (3) -18
- (4) -21

Q2 - Single Correct

Let $S_1, S_2, S_3, \dots$ and $t_1, t_2, t_3, \dots$ are two AP such that $S_1 = t_1 \neq 0$, $S_2 = 2t_2$ and $\sum_{i=1}^{10} S_i = \sum_{i=1}^{15} t_i$, then the value of $\frac{S_2 - S_1}{t_2 - t_1}$ is

- (1) $\frac{8}{3}$
- (2) $\frac{3}{2}$
- (3) $\frac{19}{8}$
- (4) 2

Q3 - Single Correct

$\frac{1}{2 \cdot 4} + \frac{1 \cdot 3}{2 \cdot 4 \cdot 6} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 8} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10} + \dots \infty$ is equal to

- (1) $\frac{1}{4}$
- (2) $\frac{1}{3}$
- (3) $\frac{1}{2}$
- (4) 1

Q4 - Single Correct

A circle of radius r is inscribed in a square. The mid-points of sides of the square have been connected by line segment and a new square resulted. The mid-point of squares sides of the resulting square were also connected by segments so that a new square was obtained and so on, then the radius of the circle inscribed in the n th square is

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(1) $\left(2^{1-n/2}\right)r$

(2) $\left(2^{3-3n/2}\right)r$

(3) $\left(2^{-n/2}\right)r$

(4) $\left(2^{5-2n/2}\right)r$

Q5 - Single Correct

$\sum_{k=1}^{360} \left(\frac{1}{k\sqrt{k+1} + (k+1)\sqrt{k}} \right)$ is the ratio of two relatively prime positive integers m and n . The value of $(m+n)$ is equal to

(1) 43

(2) 41

(3) 39

(4) 37

Q6 - Single Correct

The sum $\sum_{n=1}^{\infty} \left(\frac{n}{n^4+4} \right)$ is equal to

(1) $\frac{1}{4}$

(2) $\frac{1}{3}$

(3) $\frac{3}{8}$

(4) $\frac{1}{2}$

Q7 - Single Correct

The value of $\sum_{n=3}^{\infty} \frac{1}{n^5-5n^3+4n}$ is equal to

(1) $\frac{1}{120}$

(2) $\frac{1}{96}$

(3) $\frac{1}{24}$

(4) $\frac{1}{144}$

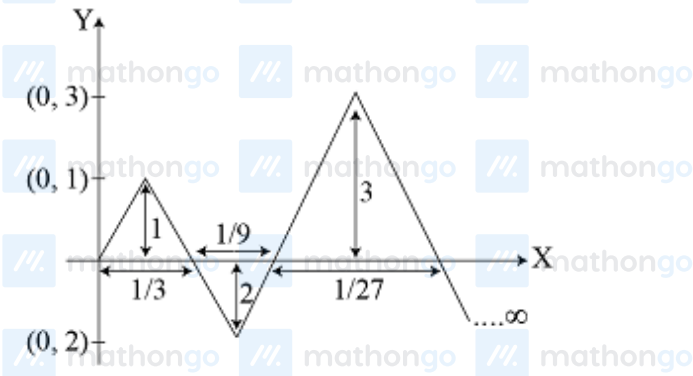
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Q8 - Single Correct

Infinite number of triangles are formed as shown in figure



If total area of these triangles is A , then $8A$ is equal to

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q9 - Single Correct

If $H_1, H_2, H_3, \dots, H_{101}$ are in HP, then $\sum_{i=1}^{100} (-1)^i \left(\frac{H_i + H_{i+1}}{H_i - H_{i+1}} \right)$ is equal to

- (1) 99
- (2) 100
- (3) 101
- (4) 102

Q10 - Multiple Correct

Let $d_1, d_2, \dots, d_k$ be all the divisors of a positive integer n in increasing order including 1 and n , if

$\sum_{i=1}^k d_i = 72$, then $\sum_{i=1}^k \left(\frac{1}{d_i} \right)$ is

- (1) $\frac{k^2}{12}$
- (2) $\frac{72}{d_k}$
- (3) $\frac{k}{72}$

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(4) None of these

Q11 - Multiple Correct

The sum of three positive numbers α, β, γ is equal to $\frac{\pi}{2}$. If $e^{\cot \alpha}, e^{\cot \beta}$ and $e^{\cot \gamma}$ form a GP. Then, which of the following hold(s) good.

(1) $\Sigma \cot \alpha = \Sigma \cot \alpha$

(2) $2 \cot \beta = \cot \alpha + \cot \gamma$

(3) $\cot \alpha \cdot \cot \gamma = 3$

(4) $4 \Sigma \cos \alpha = \Sigma \sin 2\alpha$

Q12 - Multiple Correct

If $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$, then

(1) $S = 49.5$

(2) $S = 50$

(3) $S = 49$

(4) $S = 50$

Q13 - Multiple Correct

Let a and b be positive integers. The value of xyz is 55 or $\frac{343}{55}$ according a, x, y, z and b are in AP or HP, then

(1) $a^2 + b^2 = 50$

(2) $\frac{b}{a} = 7$

(3) $a + b = 8$

(4) $a^2 + b^2 = 25$

Q14 - Multiple Correct

Which one of the following statement (s) is (are) correct?

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- (1) Sum of the reciprocals of all the n HM's inserted between a and b is equal to n times the harmonic mean between two given natural numbers a and b
- (2) Sum of the cubes of first n natural numbers is equal to square of the sum of the first n natural numbers
- (3) If $a, A_1, A_2, \dots, A_{2n}, b$ are in AP, then $\sum_{i=1}^{2n} A_i = n(a + b)$
- (4) If the first term of the GP $g_1, g_2, \dots \infty$ is unity, then the value of the common ratio of the progression such that $(4g_2 + 5g_3)$ is minimum equals $\left(\frac{2}{5}\right)$

Q15 - Multiple Correct

Let $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be AP such that $a_1 = 25, b_1 = 75$ and $a_{100} + b_{100} = 100$. Then,

- (1) the difference between successive terms in progression a is opposite the difference in progression b
- (2) $a_n + b_n = 100$ for any n
- (3) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3), \dots$ are in AP
- (4) $\sum_{r=1}^{100} (a_r + b_r) = 10000$

Q16 - Multiple Correct

Let $E = x^2 - 3x + 2, 0 \leq x \leq 2$ and $r \leq E \leq d$ for the given values of x . If

$S = 3 + (3 + d)r + (3 + 2d)r^2 + \dots \infty$, then which of the following is/are correct?

- (1) 10th term of the series is $-\frac{21}{2^{18}}$
- (2) $4r + d - 1 = 0$
- (3) Sum of infinite terms of the series equals $\left(\frac{52}{25}\right)$
- (4) $d = -2r$

Q17 - Multiple Correct

If $a, b, c \in R^+$ and $a + b + c = 8$, then

- (1) $\frac{a^2+b^2}{a+b} + \frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} \geq 8$
- (2) $\frac{a^2+b^2}{a+b} + \frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} \geq 2$
- (3) $a^2b^2 + b^2c^2 + c^2a^2 \geq 8abc$

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(4) $a^2 + b^2 + c^2 > 18$

Q18 - Multiple Correct

If $\sin(x - y)$, $\sin x$ and $\sin(x + y)$ are in HP, then $\sin x \cdot \sec \frac{y}{2}$ is

(1) 2

(2) $\sqrt{2}$

(3) $-\sqrt{2}$

(4) -2

Q19 - Multiple Correct

Let a, b, c are the first three terms of GP. If the HM of a and b is 12 and that of b and c is 36, then which of the following hold(s) good?

(1) Sum of the first term and common ratio of the GP is 11.

(2) Sum of the first five terms of GP is 948.

(3) If the value of the first term and common ratio of the given GP is taken as the first term and common difference of an AP, then its 8 th term is 29.

(4) The number 648 is one of the term of the GP.

Q20 - Multiple Correct

If $\frac{a_2 a_3}{a_1 a_4} = \frac{a_2 + a_3}{a_1 + a_4} = 3 \left(\frac{a_2 - a_3}{a_1 - a_4} \right)$, then

(1) $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \frac{1}{a_4}$ are in AP

(2) $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \frac{1}{a_4}$ are in HP

(3) a_1, a_2, a_3, a_4 are in AP

(4) a_1, a_2, a_3, a_4 are in HP

Q21 - Multiple Correct

If $x^2 + 9y^2 + 25z^2 = xyz \left(\frac{15}{x} + \frac{5}{y} + \frac{3}{z} \right)$, then

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(1) x, y, z are in AP(2) x, y, z are in HP(3) $x : y : z = 15 : 5 : 3$ (4) $x : y : z = 3 : 5 : 15$

Q22 - Multiple Correct

Consider a sequence $\{a_n\}$ with $a_1 = 2$ and $a_n = \frac{a_{n-1}^2}{a_{n-2}}$ for all $n \geq 3$, terms of the sequence being distinct. Given that, a_2 and a_5 are positive integers and $a_5 \leq 162$ then possible value(s) of a_5 can be

(1) 2

(2) 32

(3) 64

(4) 162

Q23 - Multiple Correct

If the roots of the equation $x^3 + px^2 + qx - 1 = 0$ form an increasing GP, where p and q are real, then

(1) $p + q = 0$ (2) $p \in (-3, \infty)$

(3) one of the roots is unity

(4) one root is smaller than 1 and one root is greater than 1

Q24 - Multiple Correct

If the triplets $\log a, \log b, \log c$ and $(\log a - \log 2b), (\log 2b - \log 3c), (\log 3c - \log a)$ are in arithmetic progression, then

(1) $18(a + b + c)^2 = 18(a^2 + b^2 + c^2) + ab$ (2) a, b, c are in GP(3) a, b, c are in AP(4) a, b, c cannot be the length of the sides of a triangle

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Q25 - Paragraph 1

Passage I (For Question 26, 27)

In a sequence of $(4n + 1)$ terms, the first $(2n + 1)$ terms are in AP whose common difference is 2 and the last $(2n + 1)$ terms are in GP, whose common ratio is 0.5. If the middle terms of the AP and GP are equal, then

First term of the sequence is

(1) $\frac{4n+2n \cdot 2^n}{2^n-1}$

(2) $\frac{4n-2n \cdot 2^n}{2^n-1}$

(3) $\frac{2n-n \cdot 2^n}{2^n-1}$

(4) None of these

Q26 - Paragraph 1

Middle term of the sequence is

(1) $\frac{n \cdot 2^{n+1}}{2^n-1}$

(2) $\frac{n \cdot 2^{n+2}}{2^n+1}$

(3) $n \cdot 2^n$

(4) None of these

Q27 - Paragraph 2

Passage II (For Question 28, 29)

We know that, $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = f(n)$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = g(n)$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 = h(n)$$

Greatest even natural number which divides $g(n) - f(n)$, for every $n \geq 2$ is

(1) 2

(2) 4

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(3) 6

(4) None of these

Q28 - Paragraph 2

 $f(n) + 3g(n) + h(n)$ is divisible by $1 + 2 + 3 + \dots + n$ (1) Only if $n = 1$ (2) only if n is odd(3) only if n is even(4) for all $n \in N$

Q29 - Paragraph 3

Passage III (For Question 30, 31)

Let a_m ($m = 1, 2, 3, \dots, p$) be the possible integral value of a for which the graphs of $f(x) = ax^2 + 2bx + b$ and $g(x) = 5x^2 - 3bx - a$ meets at some point for all real values of b . Let $t_r = \prod_{m=1}^p (r - a_m)$ and

$$S_n = \sum_{r=1}^n t_r, n \in N$$

The minimum possible value of a is(1) $\frac{1}{5}$ (2) $\frac{5}{26}$ (3) $\frac{3}{28}$ (4) $\frac{2}{43}$

Q30 - Paragraph 3

The value of $\sum_{r=5}^{\infty} \frac{1}{t_r}$ is equal to(1) $\frac{1}{3}$ (2) $\frac{1}{6}$ (3) $\frac{1}{15}$

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(4) $\frac{1}{18}$

Q31 - Paragraph 4

Passage IV (For Question 32, 33)

The real roots of the equation $2x^3 - 19x^2 + 57x + k = 0$ are the first three terms of a geometric progression.

The value of k is

(1) 216

(2) 100

(3) -54

(4) -108

Q32 - Paragraph 4

If the geometric progression is increasing, then the sum of its first n terms is

(1) $\left(\frac{3}{2}\right)^n - 1$

(2) $6 \left[\left(\frac{3}{2}\right)^n - 1 \right]$

(3) $4(2^n - 1)$

(4) $4 \left[\left(\frac{3}{2}\right)^n - 1 \right]$

Q33 - Paragraph 5

Passage V (For Question 34, 35)

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$ The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$, where A, B and α, β are independent of n and A is positive.

The value of $(\alpha^2 + \beta^2 + \alpha\beta)$ is

(1) 1

(2) 2

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(3) 4

(4) 5

Q34 - Paragraph 5

The quadratic equation, whose roots are α and β , is given by

(1) $x^2 - 2x - 1 = 0$

(2) $x^2 - 2x - 2 = 0$

(3) $x^2 - x - 1 = 0$

(4) None of these

Q35 - Matrix Match

Match Column I and Column II

	Column I		Column II
i.	$p^2 + 9q^2 + 25r^2 = pqr \left(\frac{15}{p} + \frac{5}{q} + \frac{3}{r} \right)$ and p, q, r are real then	p.	$p = q = r$
ii.	$\log(p+r), \log(r-p), \log(p-2q+r)$ are in A.P., then	q.	$p \neq q \neq r$
iii.	Let $f(x) = px^2 + qx + r$ be a polynomial function of second degree. If $f(1) = f(-1)$ and $f'(p), f'(q), f'(r)$ are in A.P., then	r.	p, q, r are in A.P.
iv.	Three positive numbers p, q, r satisfy (i) $p+q+r=25$ (ii) $2 < p, q, r < 18$ (iii) $2, p, q$ are in AP (iv) $q, r, 18$ are in GP, then	s.	p, q and r are in GP
		t.	p, q and r are in HP

(1)	i	ii	iii	iv
	t	t	r	q
(2)	i	ii	iii	iv
	t	r	r	q
(3)	i	ii	iii	iv
	p	t	r	q
(4)	i	ii	iii	iv
	r	t	q	t

Q36 - Integer Type

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For $a, b, c \in \mathbb{R} - \{0\}$, let $\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in AP. If α, β are the roots of the quadratic equation

$2acx^2 + 2abcx + (a+c) = 0$, then the value of $(1+\alpha)(1+\beta)$ is

Q37 - Integer Type

A pack contain n cards numbered 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the remaining number in the pack is 1224. If the smaller to the numbers on the removed cards is k , then $k - 20$, is

Q38 - Integer Type

If sum of first n terms of an AP (having positive terms) is given by $S_n = (1 + 2T_n)(1 - T_n)$ where T_n is the n th term of series, and $T_2^2 = \frac{\sqrt{a}-\sqrt{b}}{4}$ ($a, b \in \mathbb{N}$). Then $(a+b)$ is

Q39 - Integer Type

The twentieth term of an AP is $\log_{10} 20$ and the thirty second term is $\log_{10} 32$. If exactly one term of the AP is a rational number which in the lowest form is $\left(\frac{p}{q}\right)$, $p, q \in \mathbb{N}$. Then, the value of $(p+q)$ is

Q40 - Integer Type

If a, b and c are 3 positive real numbers, then minimum value of $\frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b}$ is

Q41 - Integer Type

The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 1, 3a^{-3}, a^8$ and a^{10} with $a > 0$, is

Q42 - Integer Type

Let $S_k, k = 2, 3, \dots, 100$ denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then, the value of $\frac{100^2}{100!} + \sum_{k=2}^{100} |(k^2 - 3k + 1) S_k|$, is

Q43 - Integer Type

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If a, x, b are in AP , a, y, b are in GP and a, z, b are in HP such that $x = 9z$ and $a > 0, b > 0$, where

$a \neq b, b \neq 0$ then $\frac{x^2}{y^2}$ is

Q44 - Integer Type

If $\frac{a^n + b^n}{a^{n+1} + b^{n+1}}$ is the HM between a and b where $a \neq b \neq 0$, then n is

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Answer Key

Q1 (4)

Q2 (3)

Q3 (3)

Q4 (1)

Q5 (4)

Q6 (3)

Q7 (2)

Q8 (3)

Q9 (2)

Q10 (2)

Q11 (1,2,3,4)

Q12 (1)

Q13 (1,2,3)

Q14 (2,3)

Q15 (1,2,3,4)

Q16 (1,2,3)

Q17 (1,2,3,4)

Q18 (2,3)

Q19 (1,3,4)

Q20 (1,4)

Q21 (2,3)

Q22 (2,4)

Q23 (1,3,4)

Q24 (2,4)

Q25 (2)

Q26 (1)

Q27 (2)

Q28 (4)

Q29 (2)

Q30 (4)

Q31 (3)

Q32 (4)

Q33 (2)

Q34 (3)

Q35 (1)

Q36 (1)

Q37 (5)

Q38 (6)

Q39 (9)

Q40 (3)

Q41 (8)

Q42 (3)

Q43 (9)

Q44 (0)

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